Security considerations for positioning that does not depend on external signals

Ankur Srivastava · Version 1.0 · June 2026

Protecting position information, sensing and processing in a complete geomagnetic positioning implementation

Abstract

NSATS's geomagnetic technology determines position from Earth's magnetic field without reliance on live GPS satellite signals. Removing that broadcast dependency changes which inputs a security assessment needs to examine. The assessment still follows position information through sensing, processing and the application that uses it. It covers unauthorised access to position information, disruption of the positioning task and tampering with sensors or processing. Deliberate magnetic interference and spoofed reference signals belong within that scope, alongside software integrity. This paper explains the security evaluation framework and the responsibilities of the organisation building the implementation. Positioning validation records describe agreement with a reference under recorded conditions; they do not establish security. A deployment decision requires an assessment tied to the implementation version, operating conditions, threat model and intended use of its output.

What changes when positioning does not use a live broadcast

A positioning method has dependencies that matter to the organisation using its output. NSATS's geomagnetic technology uses Earth's magnetic field, measured by sensors on the platform and processed with NSATS's proprietary noise filtering, to determine position. It supplies position information to the surrounding application without relying on live GPS satellite signals. The geomagnetic positioning page explains the distinction between a field measurement and the position calculated from it.

For this positioning function, there is no required live satellite broadcast at the receiver for an attacker to jam or spoof. That statement concerns a particular input to position calculation. The complete device still includes sensors, processing and software, and its application may exchange information with other equipment. Those parts belong in the security description of the implementation.

The title's reference to external signals concerns transmitted positioning inputs. Earth's magnetic field remains the physical input being measured. Magnetic surroundings therefore matter even when no live satellite broadcast participates in the calculation. The World Magnetic Model background distinguishes Earth's main field from other contributions to a magnetic measurement, including disturbance fields. This is physical background for assessing the measurement environment, not a description of NSATS's calculation method. 1

Ordinary environmental disturbance and deliberate magnetic interference need different assessment treatment. The first concerns conditions encountered during use; the second includes an attacker's access and intended disruption. An organisation should identify both the surroundings of the sensor and the access someone could obtain to that equipment or its environment. An assessment of ordinary operating conditions does not answer every question about deliberate interference.

Setup, operation and assessment have different inputs

An input used to prepare equipment may have a different role from one used throughout operation. A reference used only to assess position has another role again. The satellite-independent positioning page asks designers to identify those dependencies explicitly.

This distinction matters when considering a spoofed reference. A satellite receiver may supply assessment information independently of the geomagnetic position calculation. Interference with that reference then concerns the trustworthiness of the assessment record. The evaluator needs to establish which information entered the positioning function and which information was used only to judge its output.

Obtaining position locally and transmitting it elsewhere are also separate activities. A local display and a remotely monitored device use different surrounding equipment. The security scope should identify the communications used by the application and the information those connections carry, even though the positioning function does not rely on live satellite broadcasts.

The operating-environments page identifies intended directions for geomagnetic positioning. Each proposed use needs its own description of the platform, surroundings and task. An intended environment does not establish that a configuration has been assessed there, or that its security requirements are the same as those of another application.

The security evaluation framework

The Security evaluation page scopes an assessment by recording the implementation version, threat model, assessment conditions and assessment owner. A threat model describes what needs protection, the harmful actions considered and the access available to an attacker.

The framework covers the following items, in this order:

1. Unauthorised access to position information.
2. Disruption of the positioning task, including deliberate magnetic interference and spoofed reference signals.
3. Tampering with sensors or processing.

These items organise the assessment around the information and equipment used by the positioning task. The organisation applying the framework needs to connect each item to the components in its own implementation and the consequences for the application.

Unauthorised access to position information

Confidentiality concerns who can obtain position information. A position output may be used immediately in an operator's display, retained with an inspection record or passed to another part of the system. The assessment follows those uses and identifies the users or components permitted to read the information at each stage.

Access to a current position and access to stored records can serve different purposes. The application should define both. Permission to view information need not confer permission to change it; the assessment should examine those permissions separately. A user's legitimate need for one record does not describe the access required by every component handling position data.

NSATS's public run convention identifies a start point by its location code, `gps_loc_code`, and expresses positions as metres north and east of that start point. This controls what the public record presents. An implementation still needs its own rules for access to the information that its device produces, transfers and retains. Public presentation conventions do not specify those permissions.

Disruption of the positioning task

Disruption is assessed against the task that uses position. The relevant consequence may be loss of a useful output, uncertainty about an output or an interruption in the application's ability to act on it. The assessment needs the operating mode and the application's required response to interpret those consequences.

Deliberate magnetic interference belongs in the scope because the positioning input is a magnetic measurement. The assessment conditions should state the disturbance considered and the access assumed. A record of normal magnetic surroundings is useful context, but it does not establish behaviour under every deliberately introduced condition. No resistance to such interference is inferred from the presence of NSATS's proprietary noise filtering.

Spoofed reference signals require particular care during evaluation. The reference supplies information independently of the calculated position; its source and conditions must be recorded. If that information is altered, apparent agreement or disagreement may misrepresent the positioning output. The assessment therefore needs to examine the reference as well as the system being assessed.

For technical readers: An assessment-only reference must remain distinguishable from an operating input. The record should connect the reference and calculated position to the same object and observation event, explain the reference's uncertainty and identify its conditions of use. These are requirements for interpreting the evaluation, not additions to the positioning algorithm.

Tampering with sensors or processing

Tampering concerns unauthorised changes to the equipment or work that produces position information. Physical access to sensors and access through software present different assessment questions. The threat model should state the access considered so that a finding about one does not

silently stand for the other.

The installed configuration connects those questions to identifiable equipment. Sensor arrangement, mounting, orientation, processing unit, power supply and software version belong in that record. A change to the sensor, mounting or surrounding structure is recorded as a new configuration under the installation guidance.

Software integrity belongs within the processing assessment. The organisation needs to establish which software version was examined and who is permitted to change software settings. It also needs to follow changes to position records after calculation. A position can leave the expected processing component and still be altered elsewhere in the application.

The scope therefore extends to the component using the output. An unchanged position associated with the wrong object would not fulfil the intended record-keeping task. Examining the association between position and the object or observation it describes is part of assessing the complete information path.

Assessment conditions and ownership

The conditions should identify the platform, installation, motion, environment and operating mode. Setup, initialisation, restarts and configuration changes belong in the record when relevant to the work performed. Other equipment activity matters where it changes the conditions under which information is produced or used. For example, the assessment record should identify relevant communications or other software work taking place alongside positioning, and whether that activity remains constant. This allows a reviewer to distinguish changes in the surrounding system from changes to the positioning implementation. The conditions describe the arrangement actually examined.

Acceptance criteria should describe the required application behaviour before the assessment begins. A display, an inspection record and a system control use position differently. The assessment should state which use it addresses and what the application is expected to do when position information cannot support that use.

An assessment owner must be identified for the particular assessment. The commissioning organisation should define that owner’s responsibilities, the work to be performed and who will examine the resulting evidence. Recording ownership makes the scope and findings attributable; it does not imply that a named assessor or completed assessment already exists.

General security assessment guidance provides background for planning the work. NIST SP 800-115 covers planning and conducting technical security assessments, analysing findings and developing mitigation strategies. It also addresses the benefits and limitations of assessment techniques. Its role here is methodological background, not certification of an NSATS implementation. 2

Findings and their release

The framework provides for releasing results only after an assessment has been carried out. A release would identify the scope, summarise the releasable findings and state the remediation status and its date. Those details allow a reader to connect a finding to the implementation examined and to understand whether an identified issue remains open.

A statement about remediation needs to remain connected to the finding it addresses. The application version and assessed conditions also matter when interpreting whether a finding applies to a later configuration. The release should make its scope readable without disclosing proprietary positioning mechanisms or private run locations. A reviewable assessment also explains which observations were selected and why any were excluded. This lets the reviewer follow the path from the recorded work to the finding, including any judgement used to interpret the evidence.

This paper reports no completed assessment of any implementation. The fields above describe what a release would need to contain, not findings that exist.

Validation data is not security evidence

The Validation results page is where recorded runs of NSATS's own research and development (R&D) demonstration arrangement appear, each once it has been approved. The fixed definition for reading a positioning deviation is:

At each point the arrangement calculates a position afresh. Deviation is the horizontal distance between the calculated position and the reference position at that point. As a percentage, it is that distance divided by the distance moved since the previous point, multiplied by 100. The start point has no previous point and is excluded.

That definition describes a relationship between recorded positions. It does not describe who could access the information, change the processing or interfere with the reference. Validation data, including positioning deviations, is not security evidence.

Approved demonstration records relate observations to a run's setting and limitations. Public records use location codes and relative positions, with start times and elapsed seconds; a series is identified by sequence. Those conventions keep the record interpretable without presenting run locations or dates. They do not convert a positioning observation into a security finding.

The evaluation-methods page explains how an application question, independent reference and recorded conditions guide a positioning assessment. Security assessment adds the explicit threat model and examines the consequences of the harmful actions within its scope.

Applications and licensed intellectual property

The licensed positioning function supplies information that an application can display, record or use as an input to another task. The organisation building that application determines the relationship between the output and the action taken. Its security responsibilities follow that relationship through the complete implementation.

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

For geomagnetic positioning, the supplied function concerns determining position from the measured magnetic input. Software supply, documentation and updates are distinct from the surrounding application's decisions about access, installation and use of position. The licensing page sets out the current licensing position.

Installation and configuration

The installation and calibration guidance records the sensor arrangement and mounting, including orientation and distance from ferrous structures and power equipment. It also identifies the processing unit, power supply, software version, environment classes, setup sequence and acceptance check.

These details define the equipment to which the assessment applies. Mounting and nearby equipment belong alongside software version because both physical and processing changes can affect the configuration being examined. An assessment tied only to a software name would leave the sensor installation unspecified.

Calibration depends on the licensee's sensor. Where the sensor manufacturer provides a calibration procedure, that procedure applies. The guidance records whether calibration was performed and the magnetic-environment check made before positioning use. An acceptance check that confirms position output has its own purpose; the security assessment still needs the threat model and conditions relevant to that installed configuration.

Integration and the use of position

The system-integration page connects location determination to the application's display, records, controls and other information flows. The implementation needs to identify the object or observation associated with each position and preserve that association as information passes between components.

An inspection application, for example, can associate a position with a finding. The assessment then concerns access to the finding and position together, as well as whether either record can be changed. For an operator's display, the application also needs to distinguish the equipment being located from other information presented for the task. These are illustrative application requirements, not accounts of deployments.

A system control uses position within the application's task rules and operating state. Receiving an output does not itself specify the action to take. The organisation should define which component makes that decision and what information it requires, so the assessment can follow the output into its operational use.

The security and resilience page identifies source, integrity and currency as separate concerns for a location record. An unchanged record can still be too old for its intended use. The application's checks need to reflect the task's requirement for current information as well as its access rules.

Interruption and resumption

The implementation should define its response when position information cannot support the task. Depending on the application requirement, that response may involve operator review or suspending an action that needs a current position. The assessment examines the specified response in the operating mode under review.

Resumption also needs an explicit meaning. A retained position and a newly received position serve different roles after an interruption. The application may need to distinguish completed work from work awaiting action, so the assessor should follow both the position information and the associated task record.

Documentation of those behaviours gives the assessment something concrete to examine. It identifies what the application is expected to present or do, which information it needs and the conditions in which that information is usable. Security findings can then be related to the actual task rather than to the existence of a position output alone.

Post-quantum context

NSATS's Post-Quantum Security page belongs to the separate Quantum portfolio. It provides the reading route for that subject. It does not establish a security property of geomagnetic positioning or of an application using a geomagnetic position input. This paper's assessment scope remains the sensing, processing, information handling and application behaviour of the chosen positioning implementation.

The executive decision

Before deployment, the organisation should require an assessment scope that identifies the implemented configuration, its operating dependencies and the task that uses position. The scope should keep assessment-only references distinct from operating inputs and connect findings to the version examined.

Questions covered

- **Information:** Who may read position information, who may change it, and which displays, records and transfers need to be assessed?
- **Disruption:** Which deliberate magnetic interference and spoofed reference conditions are relevant to the task, and what attacker access does the threat model assume?
- **Tampering:** Which sensor, installation, software and processing changes must the assessment cover?
- **Application behaviour:** What should happen when position cannot support an action, and how should the application distinguish retained information from a new output when work resumes?
- **Assessment ownership:** Who will commission the assessment, who will carry it out, and who will judge whether its scope and evidence answer the application's requirements?
- **Findings:** What scope, releasable findings and remediation information must be available before the organisation makes its deployment decision?

The assessor the organisation names should be able to examine the relevant physical equipment, processing and application information path. The organisation commissioning the work should establish that access and remit before the assessment begins, using the Security evaluation framework to record ownership and scope.

Scientific background references

These sources provide physical and assessment-method background. Neither is evidence of NSATS's positioning performance or security.

1. NOAA National Centers for Environmental Information. *World Magnetic Model (WMM)*. The background describes the main field and other contributions to the measured magnetic field. ncei.noaa.gov/products/world-magnetic-model.
2. Scarfone, K., Souppaya, M., Cody, A. and Orebaugh, A. (2008). *Technical Guide to Information Security Testing and Assessment*. NIST Special Publication 800-115. DOI: 10.6028/NIST.SP.800-115; publication record.